NCFE Level 1 Functional Skills Qualification in Mathematics (603/5055/6)

Mark Scheme: Sample Assessment Materials

Assessment window: on demand

v1.0 Pre-standardisation

Examiner Mark Scheme guidance

Information

This guidance is intended to support NCFE examiners in the valid, reliable and consistent application of the relevant Mark Scheme version, against learner evidence generated during their external assessment.

This Mark Scheme provides:

- the total marks available for each question
- the subject content (SC) reference for each mark
- example process / methods and evidence of the types of responses expected for each mark
- (once confirmed) the pass mark for the relevant assessment version.

This Mark Scheme **must** be used for paper-based and online marking of the assessment version indicated**.**

Instructions and guidance on application

- All learners must receive the same treatment and should be marked fairly. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark Schemes should be applied positively. Learners must be rewarded for what they have shown they can do rather than penalised for things they have not done.
- Examiners should always award full marks if deserved (for example, if the answer matches the Mark Scheme). Half marks must not be awarded.
- Examiners should be prepared to award zero marks if the learner's response is not worthy of credit according to the Mark Scheme.
- The Mark Scheme is a working document and may be added to at the standardisation event to reflect valid alternative answers given by a learner.
- When in doubt regarding the application of the Mark Scheme to a learner's response, the chief examiner must be consulted.

This Mark Scheme provides the following information:

- section and activity information
- question number
- total marks available per question (top row, shaded) followed by:
    - attribution of individual marks per question
    - problem solving (PS) and underpinning skill (UPS) attribution
    - process / method or answers
    - additional or alterative evidence
    - indicative of the SC attribution
    - any additional guidance, as required.

To support the valid, reliable and consistent marking of learner evidence, the following abbreviations are applied throughout the Mark Scheme:

Annotation	Explanation and use
FT	'Follow through' marks are applied when the method uses the learner's answer from a previous step.
OE	'Or equivalent' marks are available for the justification of the answer being presented in a different form to the Mark Scheme (ie 0.5 or ½).
CAO	'Correct answer only'.
Their	'Their' refers to the learner's own derived values from a correct method or from an attempt at a correct method.
Seen	'Seen' refers to the requirement to see the stated value in the learner's response or working out.
Brackets	Indicates units are not required on final answers or for answers seen within working.
Shaded	Indicates requirements for full marks to be awarded.
Coloured SC box	**Onscreen only:** indicates where subject content reference will appear out of order in the 'Learning Outcomes' marking screen.

Version control

Mark Schemes are subject to version control. Examiners **must** ensure they have access to the latest version following each standardisation event.

Over time, Mark Schemes will incorporate additional evidence captured and confirmed during standardisation events. Any additional evidence criteria will be captured in colour-coded text applicable to the dated standardisation event.

Recording of marks

Paper-based: individual marks should be annotated in the 'Examiner' column in the learner script, added up and recorded at the end of each activity. The overall marks awarded for each learner should be clearly and legibly recorded in the grid on the back of the learner script.

Online: onscreen annotation tools (for example, ticks, crosses) should be applied to indicate application throughout the learner script, in addition to marks being recorded numerically within the corresponding 'Learning Outcomes' box, indicated by the relevant SC reference.

Annotation	Explanation and use
Tick	Used to indicate correct values / method or final answer.
Red highlight	Used to indicate where the learner has made an error in either the value used or an incorrect calculation.
BOD	Used to indicate where the learner may have made an error that has resulted in benefit of doubt being applied (ie transposition of figures but previous working clearly shows otherwise).
IMP	Implied refers to the learner's response implying correct working out used but not seen.

Paper number: SAM				Version: 1.0	
(Section A) Activity 1: underpinning skills				**(Non-calculator Test)**	
Q	**Marks**	**UPS / PS**	**Process and answer**	**Additional or alternative evidence (with guidance)**	**SC**
1 (a)	**1**	**UPS**	10 062, 100 262, 100 623, 102 630	CAO	N1b
1 (b)	**2**	**UPS**	324	Award 2 marks if correct answer given.	
	1		81 OR 4 × 81 or 4 × 9 × 9	OE Correct interpretation of square.	N6
	1		324	CAO	N7
1 (c)	**2**	**UPS**	11.86	Award 2 marks if correct answer given.	
	1		18.56 – 6.7	OE Any full correct method to find range.	H29b
	1		11.86	CAO	N11a
Activity 2: renting a flat				**(Non-calculator Test)**	
2 (a)	**3**	**PS**	See below		
(i)	1		16 × 10	OE Any correct method to work out total number of flats.	N4
	1		160 (flats)	CAO	N3a
(ii)	1		160 ÷ 10 = 16 OR 160 ÷ 16 = 10	Valid reverse check. FT Valid check using their method with their correct or incorrect answer.	N4

2 (b)	**4**	**PS**	(£)153.40	Award 4 marks if correct answer given without any working or from correct partial working. Any working seen must have correct methods and accurate figures.	
	Alternative method 1: mean first, then percentage increase				
	1		(120 + 110 + 126 + 110 + 124) ÷ 5 or 590 ÷ 5	OE Any full correct method to find mean. Accept 120 + 110 + 126 + 110 + 124 ÷ 5 Allow one error in data items.	H29a
	1		(£)118	CAO 118 implies 1st mark. Only award if from correct method for mean.	H29a
	1		Their 118 × 0.3 or 35.4 OR Their 118 × 1.3 or 153.4	OE Any full correct method to find percentage of amount or percentage increase of amount. FT Their 118 from an attempt to find mean. Allow use of any value in range. [110, 126]	N14
	1		(£)153.40	FT Their 153.4 from correct methods for finding mean and percentage increase. Must have 2dp.	N11b
	Alternative method 2: percentage increase first, then mean				
	1		120 × 1.3 or 110 × 1.3 or 126 × 1.3 or 124 × 1.3 OR 120 × 0.3 or 110 × 0.3 or 126 × 0.3 or 124 × 0.3	OE Any full correct method to find percentage or percentage increase for any value from the table.	N14
	1		156 or 143 or 163.8 or 161.2 OR 36 or 33 or 37.8 or 37.2	CAO Any correct percentage increase or percentage of value from table. Any correct value implies 1st mark.	N11b
	1		(Their 156 + their 143 + their 163.8 + their 143 + their 161.2) ÷ 5 or 767 ÷ 5 or 153.4	OE Any full correct method to find mean. FT Their values from an attempt to find percentage increase. Accept 156 + 143 + 163.8 + 143 + 161.2 ÷ 5	H29a

	1		(£)153.40	FT Their 153.4 from correct method to find percentage increase and mean. Must have 2dp.	H29a
	Alternative method 3: total, then percentage then mean				
	1		(120 + 110 + 126 + 110 + 124) or 590	OE Any full correct method to find sum.	H29a
	1		Their 590 × 0.3 OR their 590 × 1.3	OE Any full correct method to find percentage of amount or percentage increase of amount. FT Their 590 from an attempt to find total.	N14
	1		177 OR 767	CAO	N11b
	1		(£)153.40	FT Their 767 from an attempt to find sum then percentage increase. Must have 2dp. 153.4 from 767 ÷ 5	H29a
2 (c)	**3**	**PS**	See below		
	1		22.5	CAO Rounded value.	N15
	1		450 ÷ their 22.5	OE Any correct method to find width. FT Their 22.5 **only if** 22.4 or 22 or 22.47 used and **only if 1st mark not awarded.**	M22a
	1		20 (m)	FT Correct answer to 450 ÷ their 22.5 only **if 22.4** or 22 used and **only if 1st mark not awarded.** 20.08(928…) from 450 ÷ their 22.4 20.45(454…) from 450 ÷ their 22 Non-terminating decimals are not expected at level 1 but award mark if seen. Accept these values rounded or truncated from correct working seen to at least 1 dp. Do not accept rounding of 20.026(702…) from accurate calculation seen. Award this mark only for 20 seen without working.	N11b

<table>
<tr><th colspan="6">(Section B) Activity 3: underpinning skills (Calculator Test)</th></tr>
<tr><th>Q</th><th>Marks</th><th>UPS / PS</th><th>Process and answer</th><th>Additional or alternative evidence (with guidance)</th><th>SC</th></tr>
<tr><td>3 (a)</td><td>1</td><td>UPS</td><td>6 (%)</td><td>CAO</td><td>N16</td></tr>
<tr><td>3 (b)</td><td>1</td><td>UPS</td><td>±17 (°C)</td><td>CAO</td><td>N2</td></tr>
<tr><td rowspan="4">3 (c)</td><td>2</td><td>UPS</td><td>See below</td><td>Award 2 marks for fully correct frequency table.</td><td></td></tr>
<tr><td>1</td><td></td><td>3 frequencies correct
OR
4 frequencies with 2 frequencies correct which sum to 20</td><td>20 does not need to be seen in total row.</td><td>H28a</td></tr>
<tr><td>1</td><td></td><td>Fully correct table.</td><td>20 needs to be seen in total row.</td><td>H28a</td></tr>
<tr><td colspan="5">Additional guidance: 3, 6, 7, 8, 8, 9, 11, 12, 13, 16, 17, 18, 19, 22, 22, 23, 25, 27, 31, 36

<table>
<tr><th>Group</th><th>Frequency</th></tr>
<tr><td>1 – 10</td><td>6</td></tr>
<tr><td>11 – 20</td><td>7</td></tr>
<tr><td>21 – 30</td><td>5</td></tr>
<tr><td>31 – 40</td><td>2</td></tr>
<tr><td>Total</td><td>20</td></tr>
</table></td></tr>
<tr><td>3 (d)</td><td>1</td><td>UPS</td><td>730 904</td><td>CAO</td><td>N1a</td></tr>
</table>

Activity 4: part-time job				(Calculator Test)	
4 (a)	**3**	**PS**	56 (deliveries)	Award 3 marks if correct answer given without any working or from partial working. Any working seen must have correct methods and accurate figures.	
	1		16 ÷ 8 or 2 (points per minor gridline) OR 16 ÷ 1.6 or 10 (points per major gridline) OR 1.6 AND 5.6 or 8 AND 28 OR $\frac{8}{28}$ or $\frac{1.6}{5.6}$ or 0.285(714…) or 0.29 or 28.5(714…)(%) or 29(%) Value for Friday 8, Saturday 22 or Sunday 10 (day names not required).	OE Any correct method to find a relationship between the planned number of deliveries on Thursday and the number of minor or major gridlines OR State the number of gridlines for Thursday and in total OR state proportion of planned deliveries. 28 from 8 + 4 + 11 + 5 is the total number of minor gridlines. 5.6 from 1.6 + 0.8 + 2.2 + 1 is total number of major gridlines. OE Fraction. Infinite decimals are not expected at Level 1 but award mark if seen. OE Method of recognising that 8 out of 28 deliveries were planned for Thursday.	H27c
	1		Their 28 × their 2 OR their 5.6 × their 10 OR their 1.6 × their 10 + 0.8 × their 10 + 2.2 × their 10 + 1 × their 10 or 16 + 8 + 22 + 10 OR 16 × their 28 ÷ their 8 or 16 × their 5.6 ÷ their 1.6 or 16 ÷ their 0.285(714…)	OE Any full correct method to work out total number of deliveries. FT Their 2 or their 10 or their $\frac{8}{28}$ or $\frac{1.6}{5.6}$ or 0.285(714…) from correct method. FT Their 28 from correct method to add their numbers of minor gridlines for each day. FT Their 5.6 from correct method to add their numbers of major gridlines for each of day. FT Their values for total number of minor or major gridlines for each day. Applying reverse fraction or percentage method is not expected at Level 1 but award mark if seen. Accept use of 0.29 or any truncated or rounded value to at least 2 dp. Their total values for Friday 8, Saturday 22 and Sunday 10. **Day** | **Number of deliveries** Friday | 8	H27c

				Saturday 22 Sunday 10	
	1		56 (deliveries)	CAO	H27c
4 (b)	**5**	**PS**	See below	Award 5 marks if correct answer given without any working or from correct partial working. Any working seen must have correct methods and accurate figures.	
	Alternative method 1: actual distance first, then formula				
	1		24 x 500 or 12 000 m	OE Any correct method to apply scale.	M21
	1		12 (km)	CAO Converted value. 12 implies 1st mark.	M20a
	1		Their 12 ÷ 15 or 0.8 (hrs) OR 1.25 – 0.5 or 0.75 (hrs)	OE Any correct method to apply one step of formula as a forward or reverse process. FT Their 12 from an attempt to apply scale and to convert (if method seen) 0.8 implies 1st two marks.	N5
	1		Their 12 ÷ 15 + 0.5 or their 0.8 + 0.5 or 1.3 (hrs) OR their 0.75 × 15 or (1.25 – 0.5) × 15 or 11.25 (km) OR their 12 ÷ 15 or 0.8 (hrs) AND 1.25 – 0.5 or 0.75 (hrs)	OE Any full correct method to apply two steps of formula forwards or in reverse or to apply one step forwards and one step in reverse. FT Their 12 or their 0.8 from an attempt to apply scale and to convert (if method seen). 1.3 or 0.8 and 0.75 implies 1st three marks. 11.25 implies 3rd mark.	N5
	1		No AND 1.3 (hrs) OR No AND 11.25 (km) and 12 (km) OR No AND 11 250 (m) and 12 000 (m) OR No AND 0.8 (hrs) and 0.75 (hrs)	OE No supported by correct working.	N10
	Alternative method 2: actual distance first, then compare speed				
	1		24 x 500 or 12 000 m	OE Any correct method to apply scale.	M21
	1		12 (km)	CAO Converted value. 12 implies 1st mark.	M20a
	1		1.25 – 0.5 or 0.75 (hrs)	OE Any correct method to apply one step of formula as a forward or reverse process. FT Their 12 from an attempt to apply scale and to convert (if method seen).	N5
	1		Their 12 ÷ their 0.75 or 16 (km/h)	OE Any correct method to calculate speed. FT Their 12 from an attempt to apply scale and their 0.75 from a correct	N5

				method to apply one step of formula in reverse. 16 implies 1st three marks.	
	1		No AND 16 (km/h) (compared to 15 km/h)	OE No supported by correct working.	N10
	Alternative method 3: apply formula in reverse, then compare scaled distances				
	1		1.25 – 0.5 or 0.75 (hrs)	OE Any correct method to apply one step of formula as a reverse process.	N5
	1		Their 0.75 × 15 or (1.25 – 0.5) × 15 or 11.25 (km)	OE Any full correct method to apply two steps of formula in reverse. 11.25 implies 1st mark.	N5
	1		11 250 (m)	CAO Converted value. 11 250 implies 1st two marks.	M20a
	1		Their 11 250 ÷ 500 or 22(.5) (cm) OR their 11 250 ÷ 24 or 468(.75) (m) OR 24 × 500 ÷ their 11 250 or 1.06(666…) (cm)	OE Any correct method to apply scale. FT Their 11 250 from an attempt to apply formula in reverse and to convert (if method seen). Infinite decimals are not expected at level 1 but award mark if seen. Any correct value implies 1st three marks. Accept any rounding or truncation of values. 22(.5) or 468(.75) or 1.06(666…) implies 1st three marks.	M21
	1		No AND 22(.5) (cm) (compared with 24 cm) OR No AND 468(.75) (m) (compared with 500 m) OR No AND 1.06(666…) (cm) (compared with 1 cm)	OE No supported by correct working.	N10
4 (c)	**3**	**PS**	(£)97	Award 3 marks if correct answer given without any working or from correct partial working. Any working seen must have correct methods and accurate figures.	
	1		480 × 0.05 × 3 or (£)72 OR 480 + (480 × 0.05 × 3) or (£)552	OE Any full correct method to work out interest or total amount in savings account.	M18
	1		649 – (480 + (480 × 0.05 × 3)) or 649 – (480 + their 72) or 649 – their 552	OE Any full correct method to work out how much extra is needed.	M18
	1		(£)97	CAO	M18

Activity 5: underpinning skills				**(Calculator Test)**	
5 (a)	**1**	**UPS**	136(°)	Allow ± 2° tolerance.	M26b
5 (b)	**1**	**UPS**	$\frac{6}{20}$ or $\frac{3}{10}$	CAO	H31
5 (c)	**2**	**UPS**	918 (cm^3)	Award 2 marks if correct answer given.	
	1		17 × 6 × 9	OE Any correct method to find volume.	M23
	1		918 (cm^3)	CAO	M23
5 (d)	**1**	**UPS**	B	CAO	M25b
Activity 6: recycling business				**(Calculator Test)**	
6 (a)	**3**	**PS**	26 250 (bottles)	Award 3 marks if correct answer given without any working or from correct partial working. Any working seen must have correct methods and accurate figures.	
	1		$\frac{3}{8}$ OR $\frac{5}{8}$	CAO May be seen or implied in subsequent working. Allow eg 3 ÷ 8 or 3 out of 8 or 3 over 8 or 0.375 or 37.5(%) seen or implied.	N8a
	1		42 000 × 3 ÷ 8 or 15 750 OR 42 000 × 5 ÷ 8	OE Any correct method to find number of bottles used to make fabric or bags.	N9
	1		26 250 (bottles)	CAO	N9
6 (b)	**4**	**PS**	See below		
	1		81 000 (g) OR 3.6 kg	CAO Correct converted value. May be seen or implied in subsequent working.	M20b
	1		Their 81 000 ÷ 3600 OR 82 ÷ their 3.6	OE Any correct method to find number of blankets. FT Their converted value from an attempt if seen.	N17
	1		22.5 (blankets)	CAO	N17
	1		22 (blankets)	FT Their answer rounded down from an attempt to convert and an attempt to apply proportion. Their answer must come from a decimal.	N12

6 (c)	**4**	**PS**	See below		
	1		At least 2 angles correct.	May be implied by pie chart with 2 correct sectors. Two of 216º, 54º, 18º, 72º correctly implied by pie chart with 2 correct sectors (12, 3, 1, 4)	H27b
	1		All angles correct.	216º, 54º, 18º, 72º correctly implied by pie chart with correct sectors (12, 3, 1, 4)	H27b
	1		At least 2 correct sectors.	FT Their angles only if sum to 360º Allow ±2º tolerance.	H27b
	1		All sectors correct with a correct key or correct labels.	Allow ±2º tolerance.	H27b

Additional guidance
216º, 54º, 18º, 72º
Sectors: 12, 3, 1, 4



Activity 7: planning a party				(Calculator Test)	
7 (a)	**3**	**PS**	See below	Award 3 marks if correct answer given without any working or from partial working. Any working seen must have correct methods and accurate figures.	
	Alternative method 1: convert times to consistent units				
	1		1.75 (hrs) or 1 (hr) 45 (mins) OR 210 (mins) or 3 (hrs) 30 (mins) or 3.5 (hrs) OR 30 (mins) or 0.5 (hrs) OR 315 (mins) or 5.25 (hrs) or 5 (hr) 15 (mins) OR 330 (mins) or 5.5 (hrs) or 5(hrs) 30 (mins)	OE Any 1 valid time conversion or total time required or total time available. Do not award mark for incorrect use of decimal eg 5.3 315 (mins) from 105 + 210 330 (mins) from 6.15 pm to 11.45 pm	M20e
	1		315 (mins) AND 330 (mins) OR 5.25 (hrs) AND 5.5 (hrs) OR 5 (hrs) 15 (mins) AND 5 (hrs) 30 (mins) OR 11.30 (pm) OR 6.30 (pm)	OE Any comparable correct times in consistent format or correct start time or end time. Do not award mark for incorrect use of decimal eg 5.3 Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 11:30 (pm) Correct time(s) implies 1st mark.	M20e
	1		Yes AND 315 (mins) and 330 (mins) OR Yes AND 5.25 (hrs) and 5.5 (hrs) OR Yes AND 5 (hrs) 15 (mins) and 5 (hrs) 30 (mins) OR Yes AND 11.30 (pm) OR Yes AND 6.30 (pm)	OE Yes supported by correct working. FT Correct decision from their time(s). Accept comparable values in consistent units. Do not award mark for incorrect use of decimal eg 5.3 Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 11:30 (pm)	M20e

	Alternative method 2: add or subtract times				
	1		8 (pm) OR 9.45 (pm) OR 6.45 (pm) OR 10 (pm) OR 8.15 (pm)	OE Correct answer to adding one time onto 6.15 (pm) or subtracting one time from 11.45 (pm) 8 (pm) from 6.15 (pm) + 105 (mins) 9.45 (pm) from 6.15 (pm) + $3\frac{1}{2}$ (hrs) 6.45 (pm) from 6.15 (pm) + $\frac{1}{2}$ (hrs) 10 (pm) from 11.45 (pm) – 105 (mins) 8.15 (pm) from 11.45 (pm) – $3\frac{1}{2}$ (hrs) 10.15 (pm) from 11.45 (pm) – $\frac{1}{2}$ (hrs) Do not award mark for incorrect use of decimal eg 8.25(pm) Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 21:45 or 9:45 (pm) Allow this mark for adding one time correctly onto an incorrect time from adding first time onto 6.15 (pm) or for subtracting one time correctly from an incorrect time from subtracting first time from 11.45 (pm) eg 6.15 + 105 mins = 7.20 then 10.50 from 7.20 + $3\frac{1}{2}$ hrs	M20e
	1		6.30 (pm) OR 11.30 (pm) OR 8 (pm) and 8.15 (pm) OR 9.45 (pm) and 10 (pm)	OE Correct required start or end time OR correct comparative times. 6.30 (pm) from 11.45 (pm) – 105 (mins) – 3.5 (hrs) 11.30 (pm) from 6.15 (pm) + 105 (mins) + $3\frac{1}{2}$ (hrs) 8 (pm) from 6.15 (pm) + 105 (mins) 8.15 (pm) from 11.45 (pm) – $3\frac{1}{2}$ (hrs) 9.45 (pm) from 6.15 (pm) + $3\frac{1}{2}$ (hrs) 10 (pm) from 11.45 (pm) – 105 (mins) Do not award mark for incorrect use of decimal eg 6.5(pm) Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 21:45 or 9:45 (pm)	M20e
	1		Yes AND 6.30 (pm) OR Yes AND 11.30 (pm) OR Yes AND 8 (pm) and 8.15 (pm) OR Yes AND 9.45 (pm) and 10 (pm)	OE Yes supported by correct working. FT Correct decision from their time(s). Do not award mark for incorrect use of decimal eg 6.5(pm) Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 21:45 or 9:45 (pm)	M20e

	Alternative method 3: adding or subtracting one or two times and comparing difference				
	1		8 (pm) OR 9.45 (pm) OR 6.45 (pm) OR 10 (pm) OR 8.15 (pm) OR 11.15 (pm)	OE Correct answer to adding one time onto 6.15 (pm) or subtracting one time from 11.45 (pm) 8 (pm) from 6.15 (pm) + 105 (mins) 9.45 (pm) from 6.15 (pm) + $3\frac{1}{2}$ (hrs) 6.45 (pm) from 6.15 (pm) + $\frac{1}{2}$ (hrs) 10 (pm) from 11.45 (pm) – 105 (mins) 8.15 (pm) from 11.45 (pm) – $3\frac{1}{2}$ (hrs) 11.15 (pm) from 11.45 (pm) – $\frac{1}{2}$ (hrs) Do not award mark for incorrect use of decimal eg 8.25(pm) Award mark for time(s) in 12-hour format with a colon or in 24-hour format eg 21:45 or 9:45 (pm) Allow this mark for adding one time correctly onto an incorrect time from adding first time onto 6.15 (pm) or for subtracting one time correctly from an incorrect time from subtracting first time from 11.45 (pm) eg 6.15 + 105 mins = 7.20 then 10.50 from 7.20 + $3\frac{1}{2}$ hrs	M20e
	1		eg 3 (hrs) 45 (mins) OR 2 (hrs) OR 5 (hrs)	Correct comparative time(s). Award mark only if from correct method if seen. eg 1 hr 45 mins from 11.45 (pm) – 8 (pm) or from 10 (pm) – 6.15 (pm) 1 hr 50 mins from 11.45 (pm) – 9.45 (pm) or from 8.15 (pm) – 6.15 (pm) 5 hrs from 11.45 (pm) – 6.45 (pm) or from 11.15 (pm) – 6.15 (pm) Accept times in different formats.	M20e
	1		Yes AND 3 (hrs) 45 (mins) and 3 (hrs) 30 (mins) OR Yes AND 2 (hrs) and 1 (hr) 45 (mins) OR Yes AND 120 (mins) (compared with 105 mins) OR Yes AND 5 (hrs) and 4 (hrs) 45 (mins) OR Yes AND 5 (hrs) and 4.7(5) (hrs) OR Yes AND 300 (mins) and 285 (mins)	OE Yes supported by correct working. FT Correct decision from their comparable times in same format. 1 hr 45 mins from 105 mins 120 mins from 2 hrs 4 hrs 45 mins OE or 285 mins from 105 mins + 3 hrs 300 mins from 5 hrs	M20e

7 (b)	**5**	**PS**	(£)202.50	Award 5 marks if correct answer given without any working or from partial working. Any working seen must have correct methods and accurate figures.	
	Alternative method 1: perimeter of half shape first, then symmetry, then proportion				
	1		4.5 – 3.2 or 1.3 (m) OR 3.2 + 2.5 + 2 + 4.5 or 12.2 (m)	OE Any correct method to find length of missing side OR to add four given lengths. OE Consistent units.	M22b
	1		3.2 + 2.5 + their 1.3 + 2 + 4.5 or 4.5 + 2 + 2.5 + 4.5 or their 12.2 + their 1.3 or 13.5 (cm)	OE Any full correct method to find perimeter of half shape. FT Their 1.3 or their 12.2 from correct methods. 13.5 implies 1st mark.	M22b
	1		Their 13.5 × 2 or 27 (m)	OE Any full correct method to find full perimeter. FT Their 13.5 from an attempt to find length of edge of half shape. 27 implies 1st two marks.	M24b
	1		Their 27 × 7.5 or 202.5	OE Any full correct method to cost of lights. FT Their 27 from an attempt to find perimeter of half shape and to apply symmetry.	N17
	1		(£)202.50	CAO Must have 2dp.	N17
	Alternative method 2: symmetry first, then perimeter, then proportion				
	1		3.2 + 3.2 or 6.4 (m) OR 2.5 + 2.5 or 5 (m) OR 2 + 2 or 4 (m) OR 4.5 + 4.5 or 9 (m)	OE Any correct method to find double the length of any given dimension. OE Consistent units.	M24b
	1		Their 9 – their 6.4 or 2.6 (m) OR their 6.4 + their 5 + their 4 + their 9 or 24.4 (m)	OE Any correct method to find length of missing side OR to add four lengths. FT Their values from an attempt to apply symmetry. OE Consistent units. 2.6 or 24.4 implies 1st mark.	M22b
	1		Their 6.4 + their 5 + their 4 + their 9 + their 2.6 or their 9 + their 4 + their 5 + their 9 or their 24.4 + their 2.6 or 27 (m)	OE Any correct method to find full perimeter. FT Their values from an attempt to apply symmetry. 27 implies 1st two marks.	M22b
	1		Their 27 × 7.5 or 202.5	OE any full correct method to cost of lights. FT Their 27 from an attempt to find perimeter of half shape and to apply symmetry.	N17
	1		(£)202.50	CAO Must have 2dp.	N17
	Alternative method 3: proportion first, then perimeter, then symmetry				
	1		3.2 × 7.5 or (£)24 OR	OE Any correct method to find cost of one edge.	N17

			2.5 × 7.5 or (£)18.75 OR 2 × 7.5 or (£)15 OR 4.5 × 7.5 or (£)33.75		
	1		Their 33.75 – their 24 or (£)9.75 OR their 24 + their 18.75 + their 15 + their 33.75 or (£)91.5(0)	OE Any correct method to find cost of missing side OR to add four costs of given lengths. FT Their values from an attempt to apply proportion.	M22b
	1		Their 24 + their 18.75 + their 9.75 + their 15 + their 33.75 or their 33.75 + their 15 + their 18.75 + their 33.75 or their 91.5 + their 9.75 or (£)101.25	OE any full correct method to find perimeter of half shape. FT Their values from an attempt to apply proportion. 101.25 implies 1st two marks.	M22b
	1		Their 101.25 × 2 or 202.5	OE Any full correct method to apply symmetry. FT Their 101.25 from an attempt to apply proportion and find cost of edges of half shape.	M24b
	1		(£)202.50	CAO Must have 2dp.	N17
	Alternative method 4: proportion first, then symmetry, then perimeter				
	1		3.2 × 7.5 or (£)24 OR 2.5 × 7.5 or (£)18.75 OR 2 × 7.5 or (£)15 OR 4.5 × 7.5 or (£)33.75	OE Any correct method to find cost of one edge.	N17
	1		Their 24 + their 24 or (£)48 OR their 18.75 + their 18.75 or (£)37.5(0) OR their 15 + their 15 or (£)30 OR their 33.75 + their 33.75 or (£)67.5(0)	OE Any correct method to find double the cost of any given dimension. FT Their values from an attempt to apply proportion. 48 or 37.5 or 30 or 67.5 implies 1st mark.	M24b
	1		Their 67.5 – their 48 or (£)19.5(0) OR their 48 + their 37.5 + their 30 + their 67.5 or (£)183	OE Any correct method to find cost of missing side OR to add four costs. FT Their values from an attempt to apply proportion and apply symmetry. 19.5 or 183 implies 1st two marks.	M22b
	1		Their 48 + their 37.5 + their 30 + their 67.5 + their 19.5 or their 67.5 + their 30 + their 37.5 + their 67.5 or their 183 + their 19.5	OE Any correct method to find full cost. FT Their values from an attempt to apply proportion and apply symmetry.	M22b
	1		(£)202.50	CAO Must have 2dp.	N17

7 (c)	**2**	**PS**	(£)220	Award 2 marks if correct answer given.	
	1		275 × 0.2 or (£)55 OR 275 × 0.8	OE Any correct method to work out discount amount or price after discount.	M19
	1		(£)220	CAO	M19
7 (d)	**3**	**PS**	9 (tables)	Award 3 marks if correct answer given without any working or from partial working. Any working seen must have correct methods and accurate figures.	
	1		90 ÷ 5 or 18 OR 18 : 90	OE Any full correct method to apply ratio.	N17
	1		Their (18 + 90) ÷ 12 or 108 ÷ 12	OE Any full correct method to find number of tables needed. FT Their 18 from correct method to apply ratio.	N17
	1		9 (tables)	CAO	N17

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